KAMALESHWARAN P

Mechanical Engineer | Design Specialist | FEA & Simulation | AI-Integrated Systems

CAD/CAM · SolidWorks · ANSYS FEA · Product Development · Embedded Systems · AI & Computer Vision

+91 7397583709 · kamaleshwaranperumal07@gmail.com · linkedin.com/in/kamaleshwaran-p · github.com/Kamaleshwaranbadboy · Coimbatore, TN

CAREER OBJECTIVE

Mechanical Engineering student at Sri Ramakrishna Institute of Technology (CGPA: 7.93/10) with a strong foundation in design — passionate about building high-quality products and systems through precision engineering and creative problem-solving. My primary strength and ambition lie in design: from CAD modelling and structural analysis to full product development and engineering optimisation.

My career goal is to secure a senior or top-level position in a well-deserving company — whether in design engineering, product development, analysis, or any other role within the mechanical domain. I am equally open to contributing across different specializations, always aiming for the highest level of impact and responsibility wherever I am placed.

Beyond core mechanical engineering, I have intentionally built cross-disciplinary skills in coding, embedded systems, IoT, and hardware development — not to switch fields, but to make myself a more versatile and future-ready engineer. As the world evolves rapidly with computers, automation, and AI, I believe a mechanical engineer who understands intelligent systems can design smarter, more impactful solutions. These skills directly strengthen my mechanical work and enable me to contribute at the intersection of engineering and technology.

PERSONAL INFORMATION

Date of Birth	17 February 2007	**Blood Group**	B+ (ve)
Nationality	Indian	**Location**	Coimbatore, Tamil Nadu, India
Languages	English (Cambridge Linguaskills B2 — Upper Intermediate Proficient) | Tamil (Native)		
Core Strengths	Mechanical Design · Leadership · Embedded Hardware · Coding · Sports Champion		
Interests	Product Design, CAD/CAM, Structural Analysis, Motorsport Engineering, Automation, AI & Embedded Systems, 3D Printing		

EDUCATION

Qualification	Institution	Result	Year
B.E. Mechanical Engineering (2nd Year — Ongoing)	Sri Ramakrishna Institute of Technology, Coimbatore	CGPA: 7.93/10	2028*
12th Boards (HSC)	Kongunadu Matric Higher Secondary School, Velagoundampatti, Namakkal	80.1%	2023
10th Boards (SSLC)	Kongunadu Matric Higher Secondary School, Velagoundampatti, Namakkal	87.6%	2021

INTERNSHIP EXPERIENCE

Mechanical Engineering Intern — Inspection & Robotic Arm Development — Sri Jayasurya Engineering, Perur, Coimbatore

15 May 2026 – 29 May 2026 (2 Weeks)

- Completed a focused industrial internship in a live manufacturing environment, applying core mechanical engineering principles to real-world production and quality challenges.
- Contributed to the design, development, and assembly of a robotic arm mechanism — gaining hands-on exposure to robotic actuation, kinematic linkage design, and mechanical fabrication workflows.
- Participated in quality inspection processes, deepening practical understanding of dimensional tolerancing, component verification techniques, and inspection instrumentation.
- Strengthened competencies in mechanical assembly, manufacturing processes, and systematic engineering problem-solving within an industry setting.

Web Development Intern — Hospital & E-Commerce Platform Development — ZyvOne Technologies, Udumalpettai, Tiruppur, Tamil Nadu

June 2026 (2 Weeks)

- Completed a personal-initiative internship at ZyvOne Technologies — a modern software company specialising in web applications, mobile apps, and AI solutions.
- Engineered a fully functional hospital management website encompassing appointment scheduling, department information architecture, and a patient-facing responsive UI.
- Developed a fashion e-commerce platform for traditional jhumka jewellery — implementing product catalogue management, category navigation, and mobile-responsive design using HTML5 and CSS3.
- Gained industry exposure to real-world software delivery pipelines, version control workflows, and agile project execution practices.

Virtual Site Development Intern — Sri Ramakrishna Institute of Technology, Coimbatore

2024 (3rd Semester Gap Period)

- Designed and delivered a comprehensive virtual campus platform for SRIT using HTML5, CSS3, and interactive digital development concepts — creating a fully navigable digital campus experience.
- Led end-to-end project execution: virtual visualisation, interactive campus mapping, multimedia integration, and digital outreach architecture.

- Demonstrated project management, cross-functional communication, and independent technical delivery in a structured institutional environment.

PROJECTS

AI-Powered Laser QR Code Marking & Monitoring System — Indian Railways

Smart India Hackathon 2025 | Aug 2025 – Present

Technologies: *OpenCV · TensorFlow · Laser Engraving · Web Development · UDM & TMS Integration*

- Developing a portable ruggedised laser engraving system to mark durable QR and Datamatrix codes on railway track fittings — enabling end-to-end lifecycle traceability across the rail infrastructure network.
- Engineering AI-based defect detection pipelines using OpenCV and TensorFlow to autonomously verify engraving quality and flag anomalies in real time, eliminating reliance on manual inspection.
- Built a dedicated web application enabling railway maintenance staff to scan codes, retrieve full lifecycle histories, and synchronise data seamlessly with UDM and TMS platforms.
- Delivering predictive analytics dashboards providing actionable insights on inspection trends, failure patterns, warranty management, and preventive maintenance scheduling.

Autonomous Rover Navigation — Caterpillar Autonomy Challenge

Shaastra, IIT Madras — Semi-Finalist | Jan 2026 – Mar 2026

Technologies: *SLAM · Monocular Vision · Embedded Control · Computer Vision · Autonomous Path Planning*

- Engineered full autonomous navigation capabilities using SLAM (Simultaneous Localisation and Mapping) and monocular vision — qualifying as a Semi-Finalist at Shaastra IIT Madras (National Level).
- Optimised computer vision algorithms for robust real-time obstacle detection, dynamic environment mapping, and reliable path planning in unstructured terrain.
- Executed complete system integration across mechanical frame design, embedded control architecture, and navigation software — delivering a cohesive field-ready autonomous rover system.
- Collaborated in a cross-disciplinary team applying structural analysis, motor selection, and hardware-software co-design principles across the rover's full development lifecycle.

AI for Assembly — TN-IMPACT 2026 Industrial Hackathon

Dassault Systèmes & Tamil Nadu Centre of Excellence for Advanced Manufacturing | 2026

Technologies: *AI Automation · Advanced Manufacturing · Assembly Intelligence · Industry 4.0*

- Developed an AI-for-Assembly solution combining AI-driven automation techniques with advanced manufacturing principles to optimise assembly line intelligence.
- Collaborated with teammates Krishnakanth J and Nirupathunga M under mentorship of Sanjana N and technical advisory of Immanuel R — representing SRIT at a nationally prominent manufacturing innovation event.
- Gained deep exposure to Industry 4.0 thinking, AI integration in assembly automation, and smart manufacturing architecture.

AI-Based Quality Inspection System — Patent Project

Filed Indian Patent, 2025

Technologies: *Computer Vision · OpenCV · TensorFlow · AI Model Development · Hardware Interfacing*

- Co-invented an AI-powered quality inspection system for industrial quality departments — designed to automate defect detection and quality assessment of mechanical components using computer vision and AI algorithms.
- Integrated mechanical domain knowledge with AI model development and hardware interfacing to build a smart, reliable inspection pipeline that reduces human error and improves production quality standards.
- Project resulted in a filed Indian Patent (2025).

Autonomous Rover System — Patent Project

Filed Indian Patent, 2025

Technologies: *SLAM · Embedded Control · Mechanical Frame Design · Motor Selection · Firmware*

- Co-designed and co-developed a fully autonomous rover integrating mechanical design, embedded control systems, and navigation algorithms.
- Handled mechanical frame design, structural analysis, motor selection, and hardware-software integration across the rover's complete system architecture.
- Project resulted in a filed Indian Patent (2025) for the rover's innovative autonomous system design.

Kart Design & Analysis — Tamil Nadu Karting Championship Season 02

Prozone Mall, Coimbatore | Role: Kart Team Lead, Design Head & Analysis Head | Title Won

Technologies: *SolidWorks · AutoCAD · ANSYS FEA · Chassis Structural Analysis · Design Optimisation*

- Led complete end-to-end kart design: 3D CAD modelling in SolidWorks and AutoCAD through full compliance verification against competition regulations and performance specifications.
- Designed full chassis geometry, suspension layout, component packaging, and ergonomics — ensuring compliance with competition regulations and performance targets.
- Conducted rigorous ANSYS FEA on the kart chassis and all structurally critical components — validating structural integrity under dynamic racing loads and driving data-informed iterative design improvements.
- Led team coordination, component selection, fabrication oversight, and competition preparation — resulting in a Title Win in the Autosports Category.

PointIQ — Real-Time Sports Scoreboard System

Hardware & Embedded Systems Project | Independent Development

Technologies: *Arduino · ESP32 · UART / I2C / SPI · Embedded C · Real-Time Data Protocols*

- Designed and developed PointIQ — a real-time sports scoreboard and score management system for live sporting events, combining embedded firmware development with hardware integration expertise.
- Implemented multi-module data communication using UART, I2C, and SPI protocols — engineering reliable component interfacing, sensor integration, and real-time data acquisition across hardware sub-systems.

Pebble Bed Heat Exchanger — CFD & Energy Storage Analysis

National Conference, Dr. N.G.P Institute of Technology

Technologies: *ANSYS Fluent · CFD · Thermal Analysis · Heat Transfer · Energy Storage Engineering*

- Conducted detailed CFD simulation using ANSYS Fluent to evaluate heat transfer coefficients, pressure drop characteristics, and thermal energy storage efficiency across multiple pebble bed geometric configurations.
- Findings presented at the National Conference on Ghost Manufacturing and Digital Supply Chain for Viksit Bharat 2047.

Shell & Tube Heat Exchanger with Parabolic Solar Collector — Performance Evaluation

National Conference, SRM Institute of Technology | Published Conference Paper

Technologies: *ANSYS · Thermal Analysis · Solar Energy Systems · Heat Transfer Enhancement*

- Performed comprehensive performance evaluation of shell and tube heat exchangers integrated with parabolic trough solar collector systems for renewable thermal energy applications.
- Analysed thermal efficiency, heat transfer enhancement strategies, and collector-exchanger coupling under varying solar irradiance conditions — published as a peer-reviewed conference paper.

RESEARCH PAPERS & PRESENTATIONS

"Performance Evaluation of Various Shell and Tubes used in Heat Exchanger with Parabolic Solar Collector"

National Conference, SRM Institute of Technology — National Level | Published Conference Paper

"Energy Stored and Computational Fluid Dynamics Analysis of Pebble Bed Heat Exchange"

Conference on Ghost Manufacturing and Digital Supply Chain for Viksit Bharat 2047, Dr. N.G.P Institute of Technology — National Level

FORSCH'2025 — Paper Presentation

Anna University Regional Campus — University Level

Engineers Day Technical Paper Presentation

Sri Ramakrishna Institute of Technology — Institutional Level

Project Presentation at Idea Lab Inauguration

SRIT — Institutional Showcase of Innovation Projects

PUBLICATIONS & PATENTS

Patent 01 *PATENT APPLIED — India, 2025*

Augmented Reality Based Intelligent Assembly Guidance and Torque Monitoring System for Industrial Valve Manufacturing

Co-developed an AR-based intelligent system providing real-time assembly guidance and torque monitoring for industrial valve manufacturing. Combined mechanical domain expertise, software integration, and hardware interfacing to improve precision, safety, and operational efficiency in standardised industrial workflows.

Patent 02 *PATENT APPLIED — India, 2025*

Autonomous Rover System — Integrated Mechanical Design, Embedded Control & Autonomous Navigation

Co-developed and co-invented an autonomous rover system integrating mechanical structural design with embedded control, navigation, and autonomous movement capabilities. Applied end-to-end engineering from frame design and fabrication to firmware programming — resulting in a filed patent for the rover's innovative system architecture.

Patent 03 *PATENT APPLIED — India, 2025*

AI-Based Quality Inspection System for Industrial Quality Departments

Co-invented an AI-powered quality inspection system for industrial quality departments, leveraging AI and computer vision to automate defect detection and quality assessment of mechanical parts. Integrated mechanical domain expertise with AI model development to deliver a smart, reliable inspection solution that reduces human error and elevates production standards.

TECHNICAL SKILLS

Mechanical Engineering — Advanced Proficiency

CAD / Design	AutoCAD, SolidWorks, Fusion 360, Solid Edge — 2D Drafting, 3D Part Modelling & Assembly Design
Simulation / FEA	ANSYS Workbench — Structural Analysis, FEA, Thermal & Performance Evaluation
CAM & Prototyping	CAD/CAM Workflow, Tolerance & Fit Design, GD&T, 3D Printing & Rapid Prototyping
Engineering Tools	MATLAB, Simulink — Engineering Computation, Modelling & Control Systems
Core Concepts	Product Design, Design Optimisation, GD&T, Manufacturing Processes, Thermodynamics, Fluid Mechanics
Skill Level	Advanced — actively applied in live competition projects, research, and patented systems

Coding & Software — Working Proficiency

Languages	C, C++, Embedded C

Web Development	HTML5, CSS3 — Front-End Development & UI Design
Dev Tools	Git & GitHub — Version Control & Collaborative Development
IDEs / Editors	Arduino IDE, Visual Studio Code
Design Software	Canva — Technical Posters, Event Creatives & Presentation Design
Skill Level	Working — applied in embedded projects, virtual site development, and IoT integrations

Hardware & Embedded Systems — Working Proficiency

Microcontrollers	Arduino (Uno, Nano, Mega), ESP32, ESP8266 — Firmware Programming & Circuit Design
IoT Systems	Sensor Integration, Actuator Control, Real-Time Data Acquisition & Wireless Communication
Hardware Skills	Circuit Design, Breadboarding, Hardware Debugging, Component Selection & PCB Interfacing
Protocols	UART, I2C, SPI, Wi-Fi, Bluetooth — Embedded Communication Protocols
Skill Level	Working — applied in rover development, IoT sensor systems, and patented hardware projects

Leadership & Soft Skills

Leadership	Team Lead, Design Head, Analysis Head, International Service Director, Event Coordination
Teamwork	Cross-functional Collaboration, Mentoring, Peer Communication, Community Engagement
Thinking	Engineering Problem Solving, Creative Innovation, First-Principles Thinking, Goal-Oriented Mindset
Work Style	Self-Motivated, Disciplined, Deadline-Driven, Performs Under Pressure, Detail-Oriented

LEADERSHIP ROLES & RESPONSIBILITIES

Design Head — ASME SRIT Student Section

American Society of Mechanical Engineers (ASME), SRIT Chapter | 2023 – Present

- Head the design vertical of the ASME SRIT student section — responsible for all visual design, technical posters, banners, and presentation materials for events and competitions.
- Contributed strongly to ASME TN Novate 2024 & 2025 — a state-level engineering innovation competition — where the team delivered a competitive performance and gained national recognition among engineering student chapters.
- Participated in ASME EFx and ASME SJCET regional engineering events, representing SRIT with technical presentations and design showcases.

Lead Member — SAE India Student Chapter

Society of Automotive Engineers (SAE), SRIT Chapter | 2023 – Present

- Serve as a lead member driving participation in automotive engineering competitions, technical events, and SAE India initiatives.
- Participated in the 7th Edition SAEISS Bicycle Design Challenge 2025-26 at Prathyusha Engineering College — competed strongly, gained valuable industry exposure, and demonstrated applied engineering skills in a competitive national-level format.
- Coordinate team efforts for SAE-related design challenges, managing cross-functional collaboration and technical documentation.

Member — IIC (Institution's Innovation Council)

Institution's Innovation Council, SRIT | 2023 – Present

- Active member contributing to innovation-driven initiatives, idea challenges, and startup ecosystem activities under the Ministry of Education's IIC framework.
- Participated in MSME Idea Hackathon — reached the finals; secured 1st Place in the Smart India Hackathon (SIH) internal round — advanced to next stage.
- Participated in Pivot Challenge — advanced to the finals, demonstrating strong problem-solving and entrepreneurial thinking.

Innovation Ambassador — HIVE (Hub for Innovation and Entrepreneurship)

Sri Ramakrishna Institute of Technology | 2023 – Present

- Serve as Innovation Ambassador at HIVE, promoting a culture of creativity, innovation, and entrepreneurship among students at SRIT.
- Facilitate workshops, mentor aspiring innovators, and help bridge the gap between student ideas and real-world entrepreneurship resources.

Kart Team Lead

Sri Ramakrishna Institute of Technology — Kart Team | 2023 – Present

- Lead the college kart team end-to-end across design planning, component selection, fabrication coordination, and competition preparation for Tamil Nadu Karting Championship Season 02.
- Manage team workflow, delegate responsibilities across design, analysis, and build sub-teams, and drive timely delivery under competition deadlines.
- Represented SRIT at Tamil Nadu Karting Championship Season 02, Prozone Mall, Coimbatore — achieved a strong competitive result and secured a title win in the autosports category.
- Responsible for overall team strategy, communication with event organizers, and ensuring regulatory compliance.

Design Head — Kart Team

Sri Ramakrishna Institute of Technology — Kart Team | 2023 – Present

- Lead all design activities including 2D drafting, 3D modelling, part design, and full assembly using AutoCAD, SolidWorks, and Fusion 360.
- Responsible for chassis geometry, component packaging, ergonomics, manufacturability, and strict compliance with competition regulations and weight targets.
- Optimised designs for performance, safety, and structural efficiency — directly contributing to the team's competitive performance.

Analysis Head — Kart Team

Sri Ramakrishna Institute of Technology — Kart Team | 2023 – Present

- Lead structural and performance analysis of all kart components using ANSYS Workbench and engineering simulation tools.
- Conducted FEA on chassis, suspension mounts, and critical load-bearing components to identify failure points and safety margins.
- Recommended design improvements based on simulation results, contributing to a measurable improvement in structural reliability and competition readiness.

International Service Director — Rotaract Club

Rotaract International, SRIT Chapter | 2023 – Present

- Serve as International Service Director, coordinating international service projects and outreach activities at chapter level.
- Organise and lead community engagement programs, international collaboration initiatives, and large-scale service events.
- Represent SRIT Rotaract in international service domains, developing strong leadership, cross-cultural communication, and coordination skills.

Member — ASPOMERS

Sri Ramakrishna Institute of Technology | 2023 – Present

- Active member of ASPOMERS, contributing to technical events, workshops, and peer learning initiatives within the Mechanical Engineering department.

CERTIFICATIONS & TECHNICAL LEARNING

MATLAB Onramp	MathWorks Official Certification — Completed
Simulink Onramp	MathWorks Official Certification — Completed
Innovations in Automation & Machine Vision	Industry Certification
Neural Networks & AI Model Creation	AI / Machine Learning Technical Certification
Innovating with GraphRAG	Advanced AI & Knowledge Graph Certification
A Path to a Welding Engineer as Career	Welding Engineering Professional Development
How to Manage Events Like a Pro	Event Management Certification
Effective Sales and Marketing Strategies	Business & Professional Skills Certification
Learnathon	Technical Learning Program — SRIT

WORKSHOPS, COMPETITIONS & TECHNICAL PARTICIPATION

Event / Competition	Organisation / Venue	Level	Result
Tamil Nadu Karting Championship S02	Prozone Mall, Coimbatore	State	Title Won
Caterpillar Autonomy Challenge	Shaastra, IIT Madras	National	Semi-Finalist
Smart India Hackathon (Internal)	SRIT — IIC	National	1st Place
MSME Idea Hackathon	MSME Ministry of India	National	Reached Finals
Statathon — Data Analytics	National Competition	National	Reached Finals
Pivot Challenge	Startup Innovation Platform	National	Reached Finals
TN-IMPACT 2026 Hackathon	Dassault Systèmes & TN CoE	National	Participant
SAE Bicycle Design Challenge 7th Ed.	Prathyusha Engineering College	National	Strong Result
ASME TN Novate 2024 & 2025	ASME Tamil Nadu	State	Advanced Rounds
ASME EFx	ASME Regional	Regional	Tech Presentation
ASME SJCET	ASME Regional	Regional	Design Showcase
FORSCH'2025 Paper Presentation	Anna University Regional Campus	University	Presenter
Vishwakarma · Brain Bolt · Kruu Grasp	Technical Symposia	Institutional	Active Participant

AWARDS & ACHIEVEMENTS

3× Patent Co-Inventor — India, 2025

Co-inventor on three filed Indian Patents (2025): AR-based industrial assembly guidance system, autonomous rover system, and AI-powered industrial quality inspection system.

National & University Level Research Presenter

Presented research papers at two national-level conferences (Dr. N.G.P Institute of Technology & SRM Institute of Technology) and one university-level event (Anna University Regional Campus).

Tamil Nadu Karting Championship — Title Winner

Led SRIT Kart Team to a title win at Tamil Nadu Karting Championship Season 02, Prozone Mall, Coimbatore — as Kart Team Lead, Design Head, and Analysis Head.

SIH Internal Round — 1st Place

Won First Place in Smart India Hackathon (SIH) Internal Round — advanced the team to the next stage of the national hackathon.

National-Level Finals — Multiple Competitions

Reached finals in multiple national-level competitions: Caterpillar Autonomy Challenge, MSME Idea Hackathon, Statathon, and Pivot Challenge — consistently performing at the top across diverse engineering and innovation challenges.

ASME & SAE — Competitive Performer

Delivered strong competitive performances at ASME TN Novate 2024 & 2025 (State Level) and SAE Bicycle Design Challenge 7th Edition (National Level) — gaining significant recognition and technical experience.

International Service Director — Rotaract International

Selected as International Service Director, Rotaract International, SRIT Chapter — recognised for leadership and service excellence.

Design Head — ASME SRIT Student Section

Appointed Design Head of ASME SRIT Student Section — recognised for design leadership within a prestigious national engineering society chapter.

Two-Time Open State Weightlifting Champion (60 kg Category)

State-level competitive champion reflecting exceptional discipline, goal-setting, and mental resilience directly transferable to high-pressure engineering environments.

PROFESSIONAL MEMBERSHIPS

Org.	Full Name	Role / Position	Period
ASME	American Society of Mechanical Engineers	Design Head, SRIT Student Section	2023–Present
SAE India	Society of Automotive Engineers	Lead Member, SRIT Student Chapter	2023–Present
IIC	Institution's Innovation Council	Active Member, SRIT	2023–Present
Rotaract	Rotaract International	International Service Director, SRIT	2023–Present
HIVE	Hub for Innovation & Entrepreneurship	Innovation Ambassador, SRIT	2023–Present
ASPOMERS	Association of Mechanical Engineers, SRIT	Active Member	2023–Present

EXTRACURRICULAR ACTIVITIES & SPORTS

- Two-Time Open State Weightlifting Champion (60 kg Category) — State-level competitive champion; reflects the same discipline, goal-setting, and mental resilience applied in engineering and leadership.
- Competitive Bodybuilder — Structured, disciplined athlete with a consistent training regimen demonstrating commitment and physical excellence.
- Basketball Player — Active team sport participant, demonstrating teamwork, communication, and on-court leadership.
- Passionate about the intersection of Mechanical Engineering, Embedded Hardware, Software, and AI — focused on building smart, real-world physical products.
- Deeply interested in Motorsport Engineering, Automation, Intelligent Systems, 3D Printing, and Emerging Technologies that bridge the physical and digital worlds.

REFERENCES

Mr. Immanuel R
Professor, Department of Mechanical Engineering
Sri Ramakrishna Institute of Technology, Coimbatore
immanual.me@srit.org
+91 9677817992

Ms. Sanjana N
Professor, Department of Computer Science & Engineering
Sri Ramakrishna Institute of Technology, Coimbatore
sanjana.cse@sritcbe.ac.in